RT 524 Digital Agriculture

MINI Project
on

QR based product usage: Client-Server Architecture

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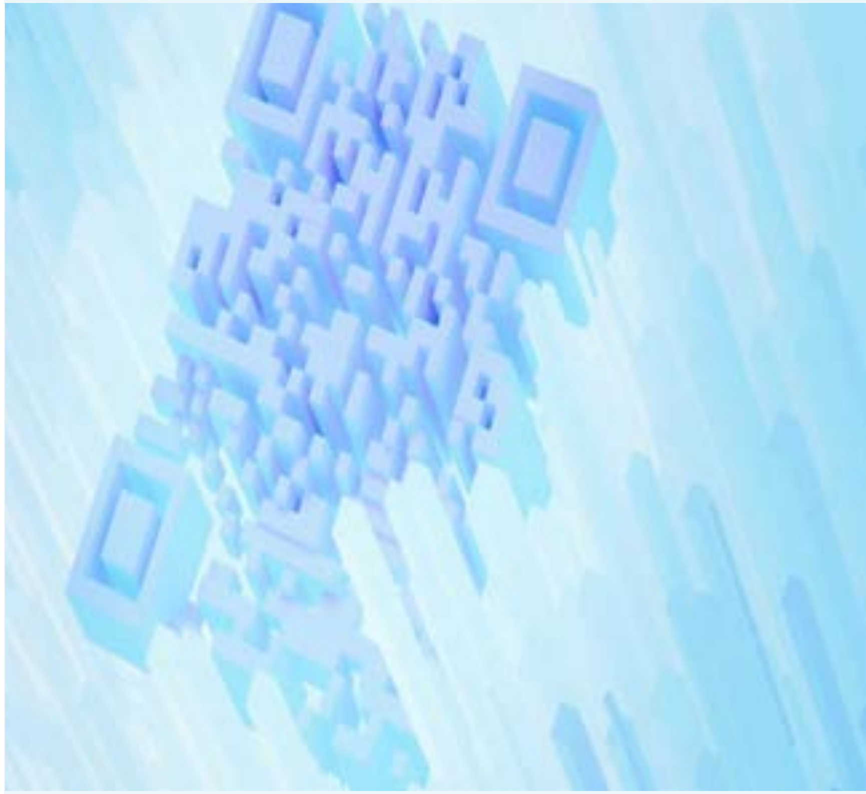
Project overview

- The QR Verification System is a web-based application developed using LAMP Sandbox environment.
- It allows users to submit QR codes either:
  - Manually (by entering QR ID)
  - By uploading QR image files
- The system stores QR details along with:
  - User information
  - Timestamp
  - Verification status
- It provides a basic framework for tracking and managing QR-based data.



Objective

- Develop a user-friendly QR-based system to record and store QR code data securely with automatic timestamping.

- Build a foundational platform that can be extended for future verification and approval processes.

Methodology

- **Technologies Used:**

LAMP Sandbox (Endor)

- Browser-based environment to run Apache, PHP, and MySQL without installation

Apache

- Manages request–response between browser and PHP

PHP

- Used for backend processing and storing QR data

MySQL (phpMyAdmin)

- Database used to store QR information with timestamp

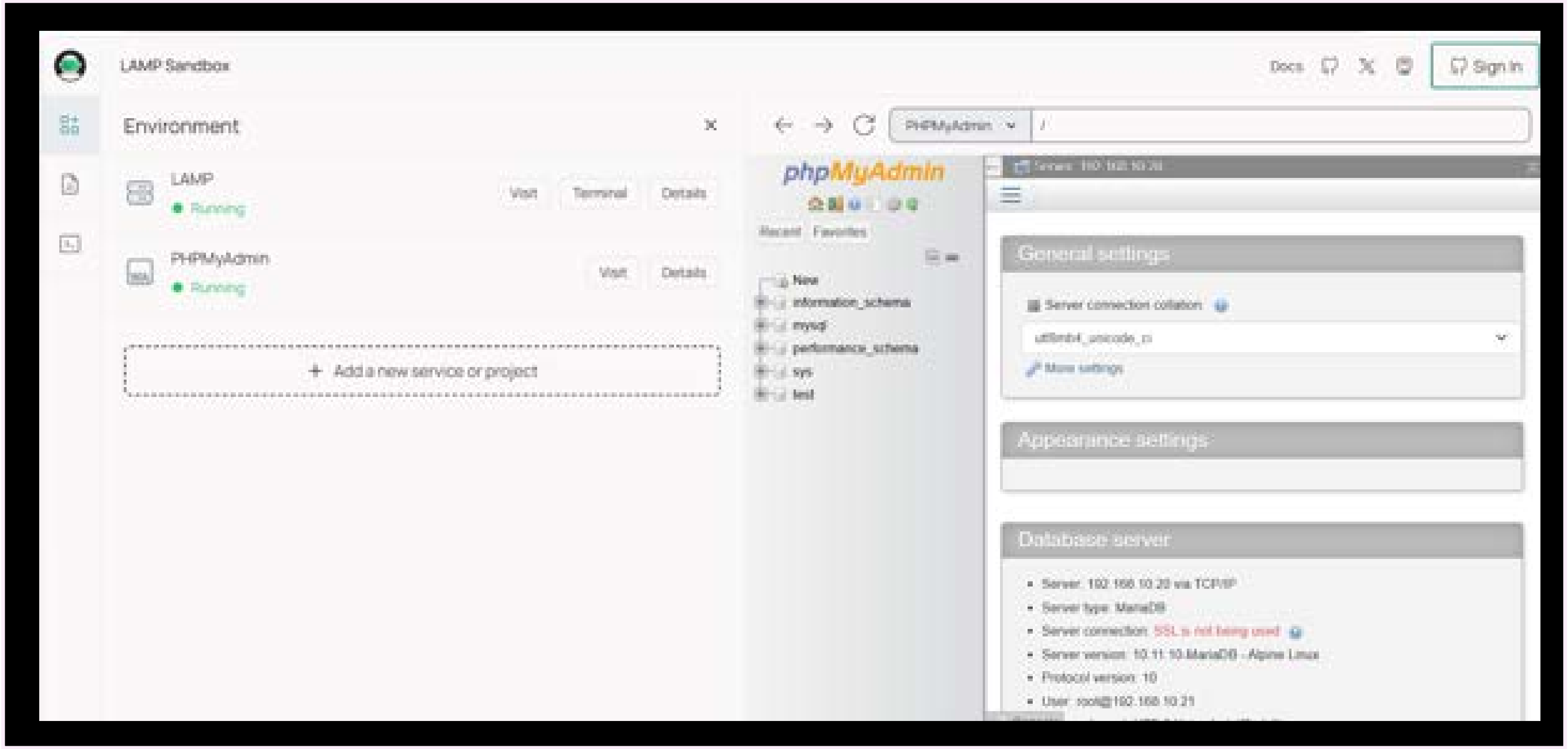
HTML & CSS

- Used to design and style the user interface

Steps of Execution:

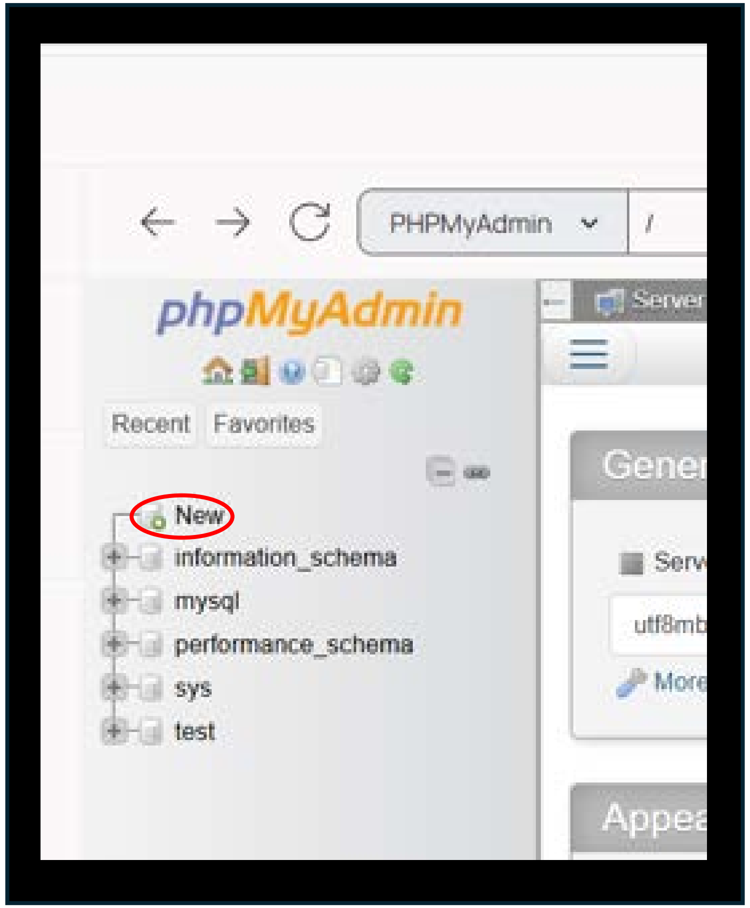
1. System set up:

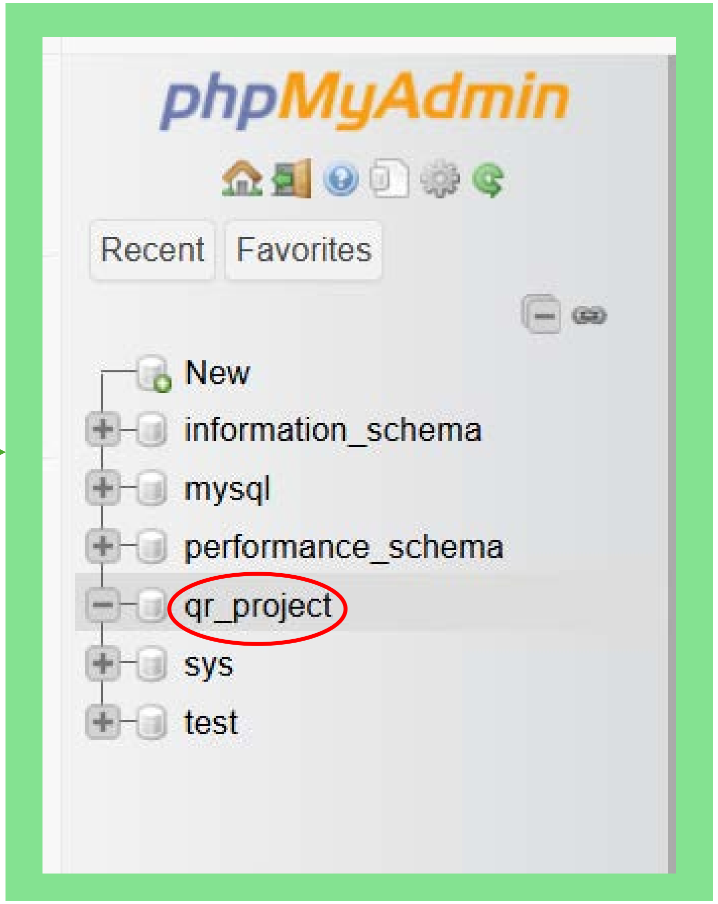
- Use LAMP sandbox(endor) as local server
- Components: Linux, Apache, MySQL, phpMyAdmin



2. Database Design:

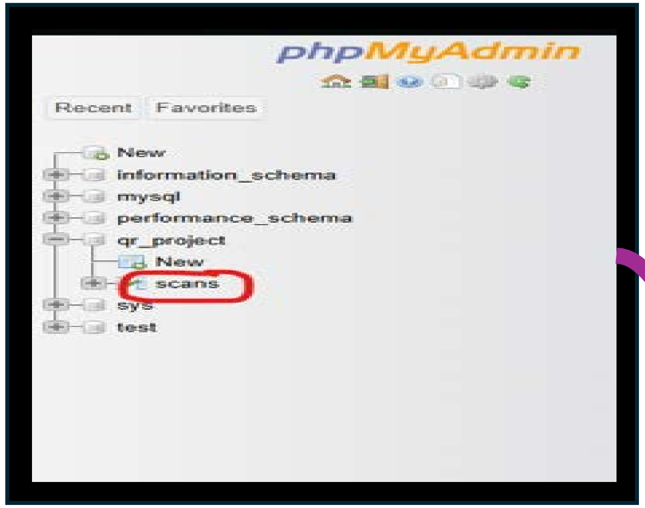
- Create database qr_project using phpMyAdmin

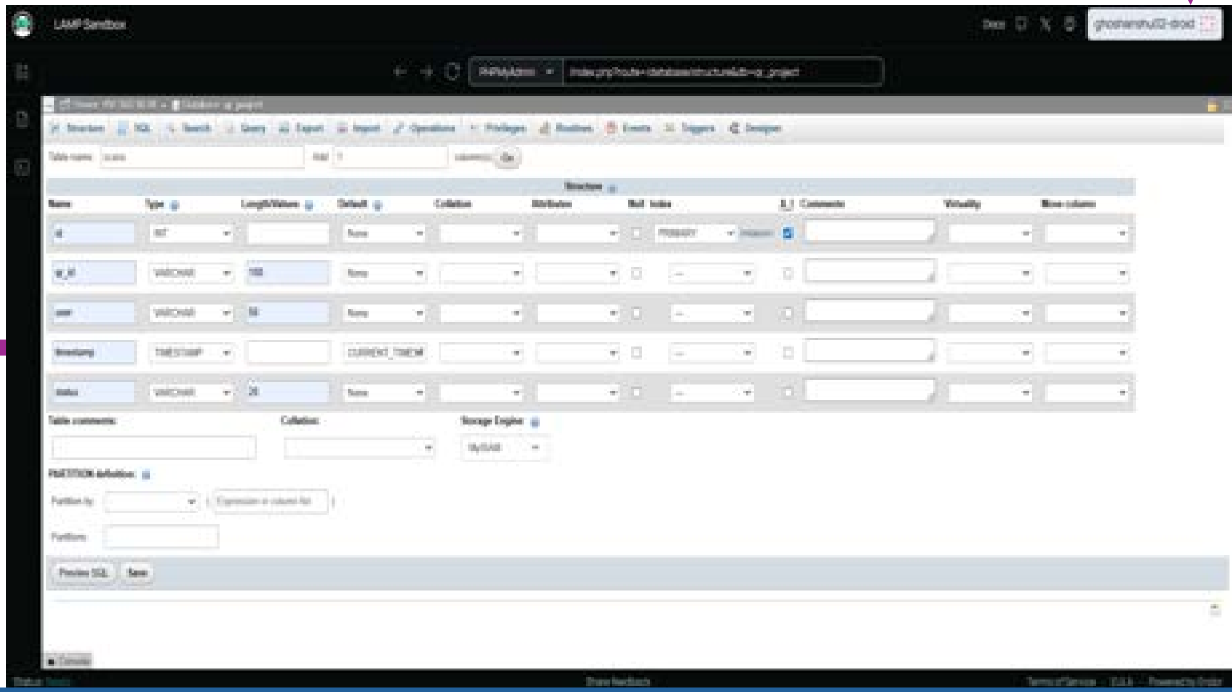


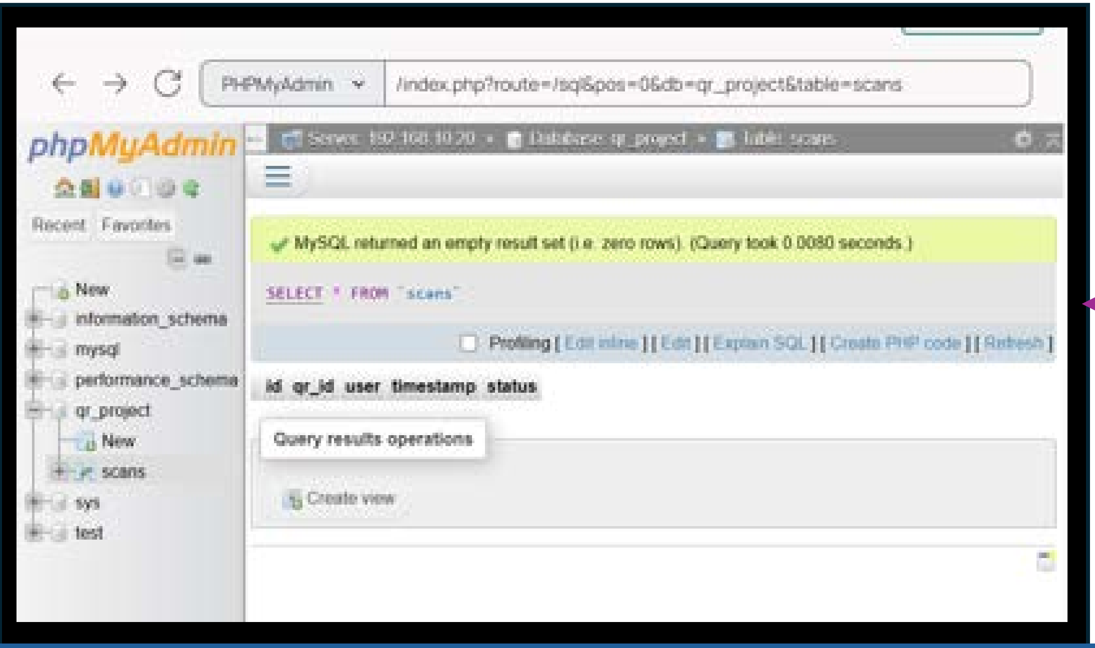


- Create table scans with fields:
  - *id (Primary Key)*
  - *qr_id*
  - *user*
  - *timestamp*
  - *status*

Name	Type	Extra
id	INT	Index (PRIMARY) + AUTO_INCREMENT (A_I)
qr_id	VARCHAR(100)	
user	VARCHAR(50)	
timestamp	TIMESTAMP	Default (CURRENT_TIMESTAMP)
status	VARCHAR(20)	

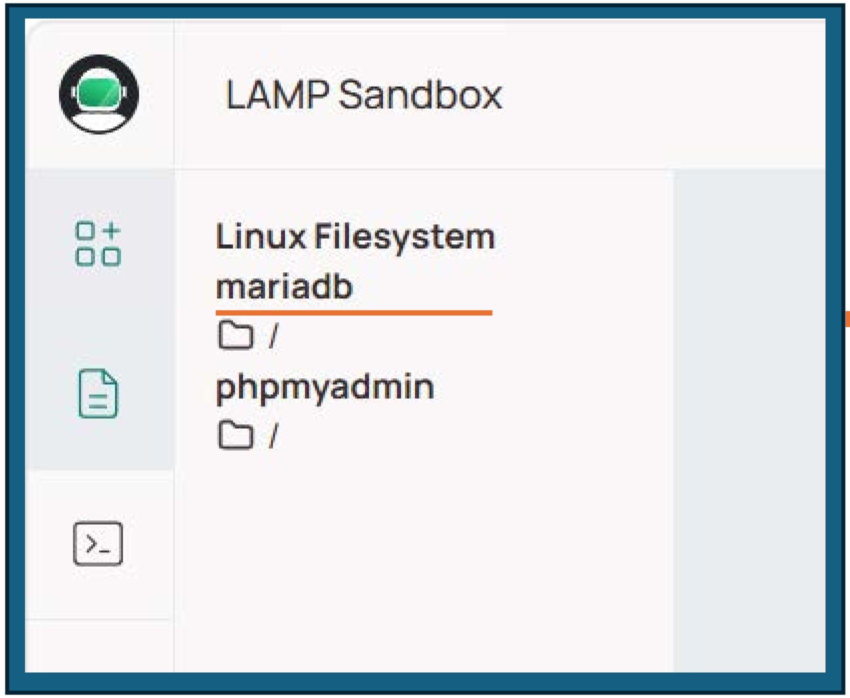


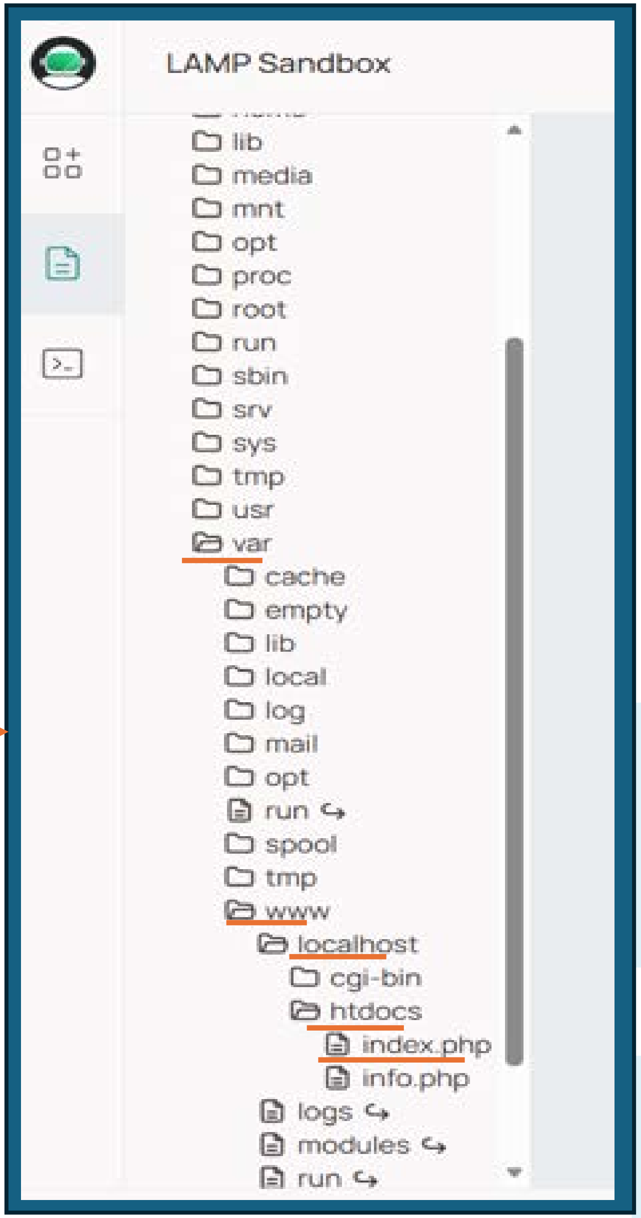




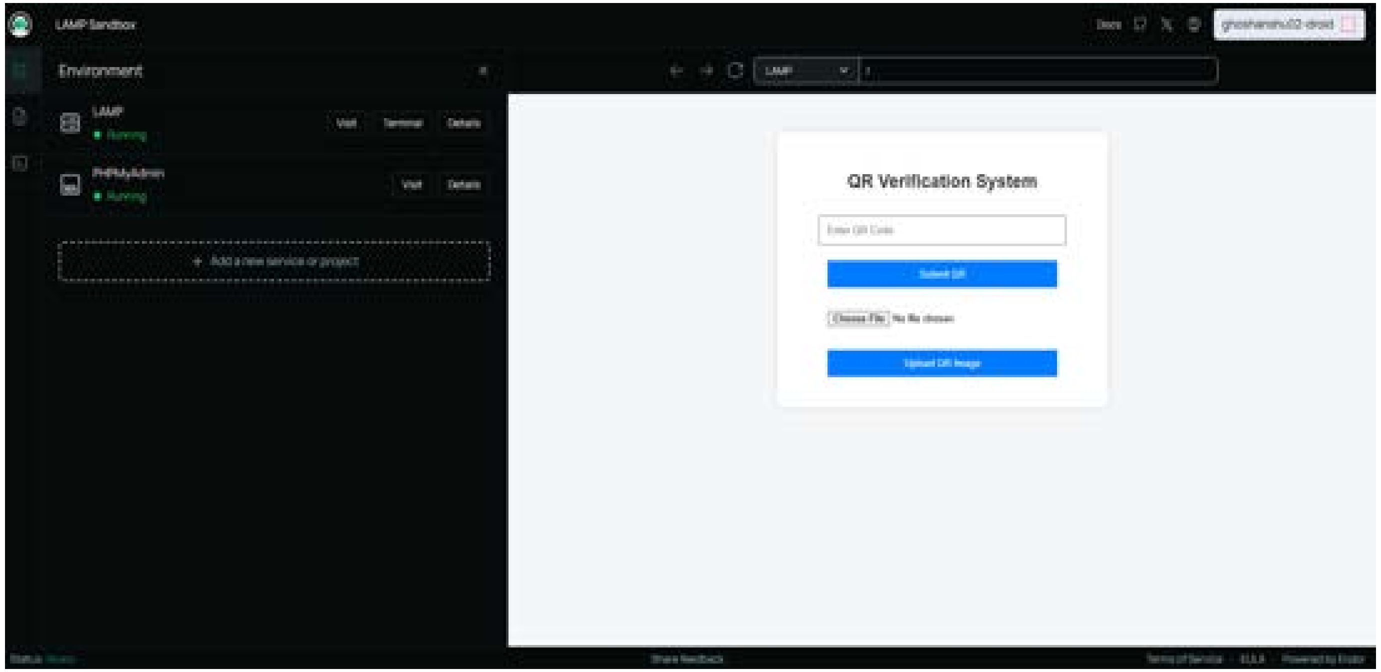
3. Develop index.php interface :

- Frontend(HTML)
- Backend(PHP)





- **System work flow**



Browser(client)

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HTML form(Frontend UI)

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HTTP request(post method)

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Apache server(LAMP)

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PHP backend processing

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My SQL database

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Response

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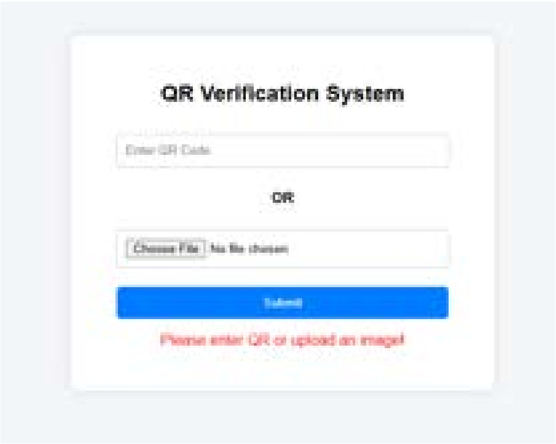
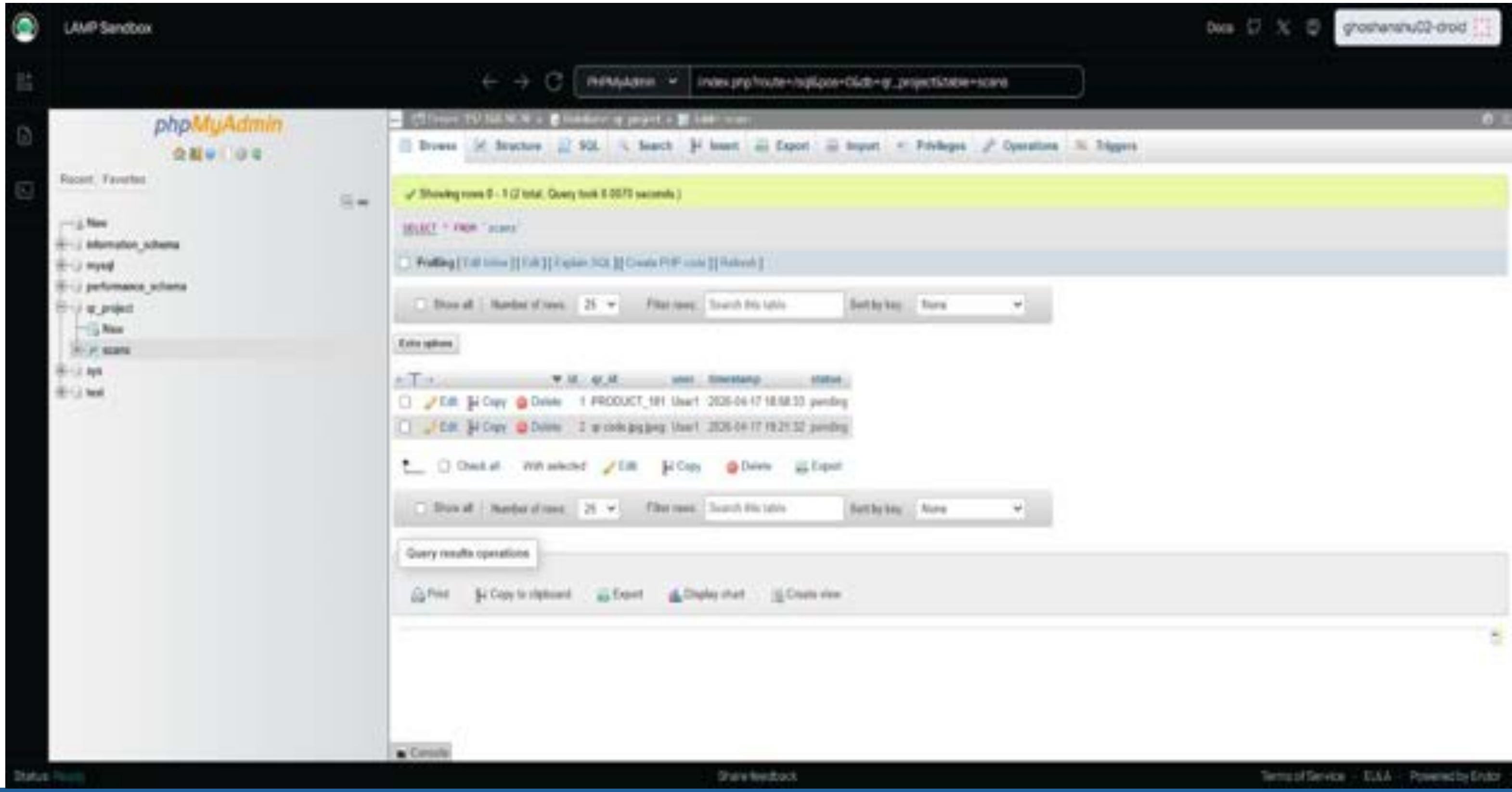
Browser display

Result

- Successfully created a working QR submission system
- Users can:
  - Enter QR codes manually
  - Upload QR image files
- Data is stored in database with:
  - Unique ID
  - Timestamp
  - Status (pending)
- System runs on local server using LAMP





Future Scope

- Add QR code scanning using mobile camera
- Implement admin panel for approval/rejection
- Improve QR image decoding instead of using filename
- Add user authentication (login system)

Conclusion

- The project demonstrates a basic implementation of a QR-based tracking system
- It successfully integrates frontend and backend using LAMP stack
- The system is functional but has scope for improvement

THANK YOU